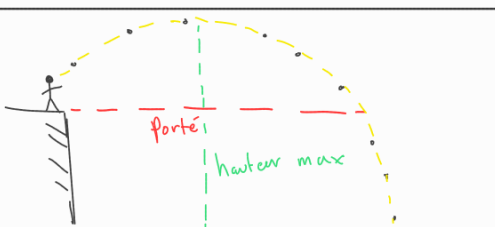
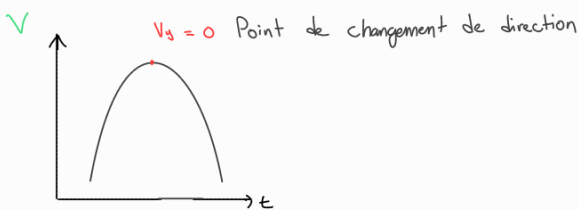
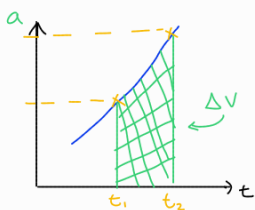


déplacement = $\rightarrow$ Vecteur
distance parcourue = $3+2=5 \rightarrow$ scalaire



MRU : Vitesse constante $\rightarrow V = Ct$
① $x_2 = x_1 + v_1(t_2 - t_1)$

MRUA : Accélération constante
① $x_2 = x_1 + v_1(t_2 - t_1) + \frac{1}{2}a(t_2 - t_1)^2$
② $v_2 = v_1 + a(t_2 - t_1)$
 $v_2^2 - v_1^2 = 2a(x_2 - x_1)$

Chute Libre (MRUA)
 $a_j = -g = -9.81 m/s^2$

① $y_2 = y_1 + v_{y1}(t_2 - t_1) - \frac{1}{2}g(t_2 - t_1)^2$

② $v_{y2} = v_{y1} - g(t_2 - t_1)$

$(v_{y2})^2 = (v_{y1})^2 - 2g(y_2 - y_1)$

Projectile * Si il ya un angle utilise v_x et v_y ...
 $v_x = \cos(\theta) \cdot v$
 $v_y = \sin(\theta) \cdot v$

$a_y = -g = -9.81 m/s^2$

Selon x (MRU)

① $x_2 = x_1 + v_{x1}(t_2 - t_1)$

Selon y (MRUA)

① $y_2 = y_1 + v_{y1}(t_2 - t_1) - \frac{1}{2}g(t_2 - t_1)^2$

② $v_{y2} = v_{y1} - g(t_2 - t_1)$

#3

	①	②	③	④
x	$x_1 = 0$	$x_2 = ?$	$x_3 = ?$	$x_4 = 3000$
v	$v_1 = 27.78$	$v_2 = 0$	$v_3 = 0$	$v_4 = 36.11$
t	$t_1 = 0$	$t_2 = ?$	$t_3 = ?$	$t_4 = 0$

$t_2 = 27.78s$
 $x_2 = 585.86m$

$x_3 = 651.97$
 $t_3 = 36.11$

Problème d'accélération constante
Selon X

MRUA selon X

① $x_2 = x_1 + v_1(t_2 - t_1) + \frac{1}{2}a(t_2 - t_1)^2$
② $v_2 = v_1 + a(t_2 - t_1)$

$x_2 - x_1 > 0$ Pas de collision

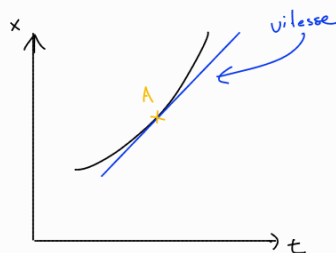
Vitesse Moyenne $[m/s]$

$\vec{V} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$

Vitesse instantanée

$V = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t}$

$v > 0$: Déplacement côté positif
 $v < 0$: " " négatif

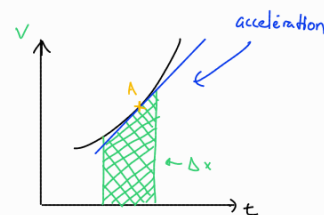


accélération moyenne $[m/s^2]$

$\vec{a} = \frac{\Delta v}{\Delta t} = \frac{v_2 - v_1}{t_2 - t_1}$

accélération instantanée

$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t}$



Trigonometrie

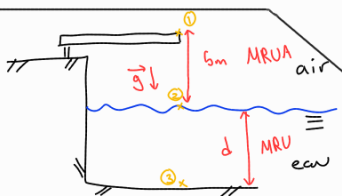
$\sin(\theta) = \frac{opp}{hyp}$

$\tan(\theta) = \frac{opp}{adj} = \frac{\sin \theta}{\cos \theta}$

$C^2 = a^2 + b^2$

$\theta = \tan^{-1}(\frac{a}{b}) = \tan^{-1}(\frac{y}{x})$

$\cos(\theta) = \frac{adj}{hyp}$



① MRUA

② MRU

③

$y_1 = d + 5$

$y_2 = d$

$y_3 = 0$

$v_{y1} = 0$

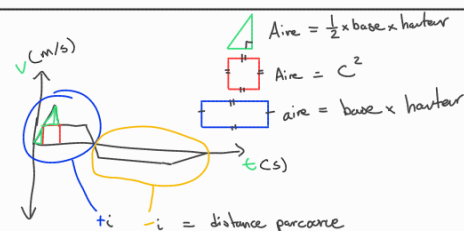
$v_{y2} = ?$

$v_{y3} = v_{y2}$

$t_1 = 0$

$t_2 = ?$

$t_3 = 5s$



Projectile

Selon x : MRU

(1) $x_2 = x_1 + v_{x1}(t_2 - t_1)$

Selon y : MRUA

(2) $y_2 = y_1 + v_{y1}(t_2 - t_1) - \frac{1}{2}g(t_2 - t_1)^2$

(3) $v_{y2} = v_{y1} - g(t_2 - t_1)$

①

②

$y_1 = 730m$

$x_1 = 0$

$v_{y1} = -(\cos 53 \cdot v_1)$

$v_{x1} = (\sin 53 \cdot v_1)$

$t_1 = 0$

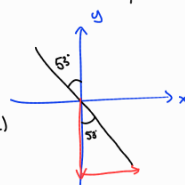
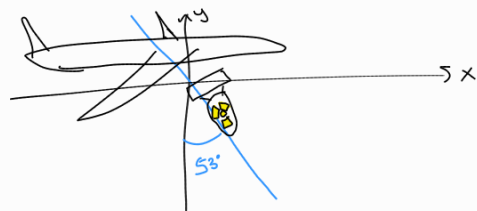
$y_2 = 0$

$x_2 = ?$

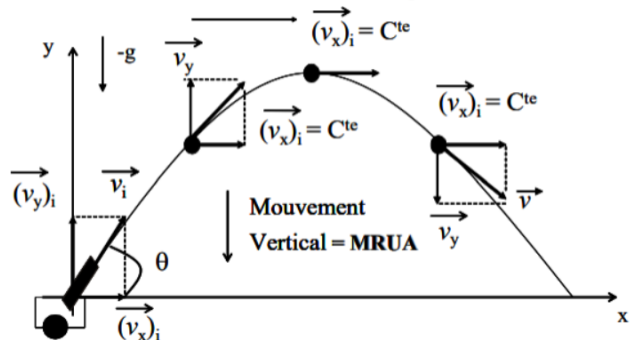
$v_{y2} = -(\cos 53 \cdot v_2)$

$v_{x2} = (\sin 53 \cdot v_2)$

$t_2 = 5$



Mouvement horizontal = MRU



MCUA

ω_0 : vitesse angulaire initiale (rad/s)
 t : temps
 θ_0 : angle initial (rad)
 $\theta(t)$: angle au temps $t \rightarrow \theta(t) = \theta_0 + \omega_0 \cdot t + \frac{1}{2} \alpha \cdot t^2$
 R : rayon du cercle
 α : accélération angulaire C^{te}
 $\omega(t)$: vitesse angulaire au temps $t \rightarrow \omega(t) = \omega_0 + \alpha \cdot t$
équation d'un MCUA
 $(\omega_2)^2 - (\omega_1)^2 = 2 \cdot \alpha \cdot (\theta_2 - \theta_1)$
 $\theta_2 = \theta_1 + \omega_1(t_2 - t_1) + \frac{1}{2} \alpha (t_2 - t_1)^2$ ①
 $\omega_2 = \omega_1 + \alpha(t_2 - t_1)$ ②

Paramètre angulaire et linéaire

$$L = \theta \cdot R \quad \theta = \frac{L}{R}$$

$$V = \omega \cdot R \quad \omega = \frac{V}{R}$$

$$a_t = \alpha \cdot R \quad \alpha = \frac{a_t}{R}$$

Système contraint

Courie / engrenage

$$\theta_1 R_1 = \theta_2 R_2$$

$$\omega_1 R_1 = \omega_2 R_2$$

$$\alpha_1 R_1 = \alpha_2 R_2$$

$$V_1 = V_2$$

$$L_1 = L_2$$

$$a_{t1} = a_{t2}$$

Roues concentrique

$$\theta_1 = \theta_2$$

$$\omega_1 = \omega_2$$

$$\alpha_1 = \alpha_2$$

$$R = 300 \text{ mm}$$

$$V_A = 15 \text{ m/s}$$

$$V_B = 20 \text{ m/s}$$

$$a_t = \alpha R = \alpha \cdot R$$

$$V_A = \omega_A \cdot R \rightarrow \omega_A = 0,05$$

$$V_B = \omega_B \cdot R \rightarrow \omega_B = 0,066667$$

$$\theta_2 = \theta_1 + \omega_1(t_2 - t_1) + \frac{1}{2} \alpha(t_2 - t_1)^2$$

$$\omega_2 = \omega_1 + \alpha(t_2 - t_1)$$

①	$\alpha = ?$	②
$\theta_1 = 0$		$\theta_2 = \frac{\pi}{2}$
$\omega_1 = 0,05$		$\omega_2 = 0,066667$
$t_1 = 0$		$t_2 = ?$

$$\frac{\pi}{2} = 0 + 0,05(t_2 - 0) + \frac{1}{2} \alpha(t_2 - 0)^2$$

$$0,066667 = 0,05 + \alpha(t_2 - 0)$$

$$t_2 = 26,9279 \text{ sec}$$

$$\alpha = 0,000619 \text{ rad/sec}^2$$

$$a_t = 0,000619 \cdot 300 = 0,186 \text{ m/s}^2$$

b)

①	$\alpha = 0,000619$	②
$\theta_1 = 0$		$\theta_2 = \frac{\pi}{2}$
$\omega_1 = 0,05$		$\omega_2 = ?$
$t_1 = 0$		$t_2 = ?$

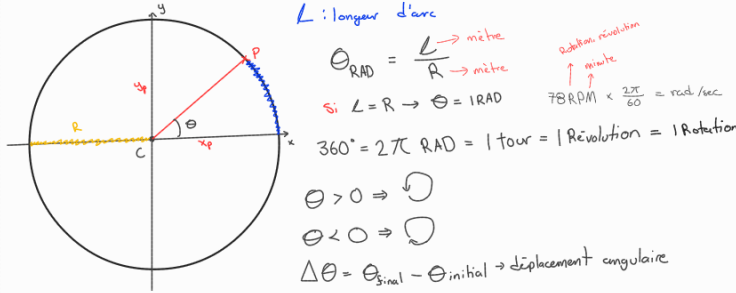
$$t_2 = 6,412 \text{ sec}$$

$$\omega_2 = 0,054 \text{ rad/sec} \Rightarrow V_2 = \omega_2 \cdot R = 0,054 \cdot 300 = 16,14 \text{ m/s}$$

$$a_c = \frac{V_2^2}{R} = \frac{(16,14)^2}{300} = 0,87372 \text{ m/s}^2$$

MCU

ω : vitesse angulaire C^{te} (rad/s)
 t : temps
 θ_0 : angle initial (rad)
 $\theta(t)$: angle au temps t
 V : vitesse tangentielle
 R : rayon du cercle
MCU
 $\theta_2 = \theta_1 + \omega(t_2 - t_1)$



MCUA

$$\theta_2 = \theta_1 + \omega_1(t_2 - t_1) + \frac{1}{2} \alpha(t_2 - t_1)^2$$

$$\omega_2 = \omega_1 + \alpha(t_2 - t_1)$$

①	$\alpha = ?$	②	③
$\theta_1 = 0$		$\theta_2 = ?$	$\theta_3 = \theta_2 + 60$
$\omega_1 = 0$		$\omega_2 = 10 \text{ RPS}$	$\omega_3 = 15 \text{ rps}$
$t_1 = 0$		$t_2 = ?$	$t_3 = t_2 + t$

$$2 \rightarrow 3$$

$$\theta_3 + 60 \cdot 2\pi = \theta_2 + (10 \cdot 2\pi)(t_3 - t_2) + \frac{1}{2} \alpha(t_3 - t_2)^2$$

a) $15 \cdot 2\pi = 10 \cdot 2\pi + \alpha(t_3 - t_2)$

$$\alpha = 6,54498 \text{ rad/s}^2$$

$$t = 4,8 \text{ s}$$

accélération angulaire moyenne

$$\bar{\alpha} = \frac{\Delta\omega}{\Delta t} = \frac{\omega_f - \omega_i}{t_f - t_i}$$

accélération angulaire instantanée

$$\alpha = \lim_{\Delta t \rightarrow 0} \frac{\Delta\omega}{\Delta t}$$

accélération tangentielle

$$[m/s^2] \quad [rad/s^2] \quad [mètre]$$

$$a_t = \alpha \cdot R$$

accélération centripète

$$a_c = V \cdot \omega \rightarrow V = \omega \cdot R$$

$$a_c = \frac{V^2}{R}$$

accélération total

$$\vec{a}_{\text{total}} = \vec{a}_t + \vec{a}_c$$

$$a_t = \frac{\Delta V}{\Delta t} = \alpha \cdot R$$

$$a_{\text{total}} = \sqrt{a_t^2 + a_c^2}$$

$\alpha = 6,54498 \text{ rad/s}^2$

①	$\alpha = 6,54498$	②
$\theta_1 = 0$		$\theta_2 = 60 \cdot 2\pi$
$\omega_1 = 0$		$\omega_2 = ?$
$t_1 = 0$		$t_2 = ?$

$$60 \cdot 2\pi = 0 + 0 \cdot (t_2 - 0) + \frac{1}{2} \alpha(t_2 - 0)^2$$

$$\omega_2 = 0 + \alpha(t_2 - 0)$$

$$\omega_2 = 70,2481 \text{ rad/s}$$

$$t_2 = 10,7 \text{ sec}$$

c)

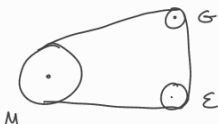
①	$\alpha = 6,54498$	②
$\theta_1 = 0$		$\theta_2 = ?$
$\omega_1 = 0$		$\omega_2 = 10 \cdot 2\pi$
$t_1 = 0$		$t_2 = ?$

$$\theta_2 = 0 + 0(t_2 - t_1) + \frac{1}{2} \alpha(t_2 - 0)^2$$

$$10 \cdot 2\pi = 0 + \alpha(t_2 - 0)$$

$$\theta_2 = 301,593 \text{ rad} \rightarrow 48 \text{ tour}$$

$$t_2 = 9,6 \text{ sec}$$



$$D_m = 20 \text{ cm} = 0,2 \text{ m}$$

$$D_g = 8 \text{ cm} = 0,08 \text{ m}$$

a) $\omega_m R_m = \omega_e R_e$

$$\omega_e = 2\omega_m$$

$$\omega_m R_m = 2\omega_m R_e$$

$$R_e = \frac{R_m}{2} = 0,05 \text{ m} = 5 \text{ cm}$$



b) MCUA $\alpha = C^{te}$

①	$\alpha = ?$	②
$\theta_1 = 0$		$\theta_2 = ?$
$\omega_1 = 3000 \text{ RPM}$		$\omega_2 = 4000 \text{ RPM}$
$t_1 = 0$		$t_2 = 4 \text{ s}$

$$\theta_{2m} = 0 + (3000 \cdot \frac{2\pi}{60})(4 - 0) + \frac{1}{2} \alpha(4 - 0)^2$$

$$(4000 \cdot \frac{2\pi}{60}) = (3000 \cdot \frac{2\pi}{60}) + \alpha(4 - 0)$$

$$\alpha = 26,18 \text{ rad/s}^2$$

$$\theta_2 = 1466,08 \text{ rad}$$

$$\text{accélération courroie} \leftarrow a_t = \alpha_m \cdot R_m = 26,18 \cdot 0,1 = 2,618 \text{ m/s}^2$$

